Hereditary Diffuse Gastric Cancer

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Genetic syndrome first reported among Maori population in New Zealand

- Autosomal dominant
- Early-onset diffuse gastric cancer ($\tilde{x}$ age = 40)
- High penetrance

CDH1 mutations and HDGC

Pathologic variants in CDH1 gene associated with HDGC

Penetrance in Populations Enriched in HDGC

Prophylactic Gastrectomy

- Stanford

90% of patients undergoing prophylactic gastrectomy harbor microscopic foci of gastric cancer in their stomach

Endoscopic Screening - NIH

A large NIH cohort study of 270 CDH1 carriers undergoing surveillance with systematic gastric mapping found only two patients developed clinically evident pT3N0 gastric carcinomas over a median 31-month follow-up, with no gastric cancer deaths.

(Asif et al. 2023)

Lifetime Risk of Cancer

Cancer endpoint	Cumulative risk to age 80	Mean age at onset	References
Advanced gastric ca (stage II), males	10.3%	47-49y	4][35][2]
Advanced gastric ca (stage II), females	**~6.5–7%**	47-49	
Gastric ca including occult pT1a, males	37-42%		4 35 2
Gastric ca including occult pT1a, females	25-33%		
Lobular breast cancer (females)	**37–55%**	51–54y	1 2 3 8

Endoscopic Screening - Cambridge

imilarly, a 16-year longitudinal cohort study demonstrated that endoscopic surveillance with systematic random biopsies can identify early lesions and inform decision-making about gastrectomy.

(Lee et al. 2023)

Endoscopic Screening

2025 study of 20 carriers undergoing long-term surveillance (mean 62 months) found no patients developed aggressive

(greater than pT1a) cancer, even among those with positive pT1a biopsies who continued surveillance.

aside: Expert Endoscopic Surveillance in CDH1 Pathogenic Variant Carriers Seems Safe, Even After Positive (pT1a) Biopsies. Journal of Medical Genetics. 2025. Benusiglio PR, Leenhardt R, Duros C, et al.New

Asif, Bilal, Amber Leila Sarvestani, Lauren A Gamble, et al. 2023. "Cancer Surveillance as an Alternative to Prophylactic Total Gastrectomy in Hereditary Diffuse Gastric Cancer: A Prospective Cohort Study." *The Lancet Oncology* 24 (4): 383–91. https://doi.org/10.1016/S1470-2045(23)00057-8.

Lee, Colin Y. C., Adriaan Olivier, Judith Honing, et al. 2023. "Endoscopic Surveillance with Systematic Random Biopsy for the Early Diagnosis of Hereditary Diffuse Gastric Cancer: A Prospective 16-Year Longitudinal Cohort Study." *The Lancet. Oncology* 24 (1): 107–16. https://doi.org/10.1016/S1470-2045(22)00700-8.